Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.5223	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	79.42908	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1471.5379	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.263478	0.263478	160.56024	0.657136	12.115

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.5223.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 79.42908.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1471.5379.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.297436.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.263478.
- Held-out runtime ms: 160.56024.
- Robustness dispersion across held-out structure families: 0.003514.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.400708, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.353572, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.307367, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.137927, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.117816, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver Evolution Suite

Condition	Feasibility Rate	Held-out Feasible Gap	Held-out Runtime (ms)	Robustness Dispersion	Notes on Objective & Runtime
Baseline Nearest Neighbor Constructive	100%	0.2734	43.5	0.0028	Fastest runtime, moderate gap
Baseline Clarke-Wright Savings	100%	**0.0738**	79.4	0.0078	Best objective gap, moderate runtime
Baseline Regret Insertion + Local Search	100%	0.4664	1471.5	0.0869	Worst objective gap, slowest runtime
Evolved Solver	100%	0.2635	160.6	0.0035	Moderate gap and runtime, good robustness, code novelty

Feasibility

- All conditions, including the evolved solver, achieved perfect feasibility (100%) on held-out instances.

Objective Gap

- Clarke-Wright Savings baseline clearly outperforms all others with the lowest feasible gap (0.0738).
- The evolved solver has a feasible gap (0.2635) slightly better than Nearest Neighbor (0.2734) but much better than Regret Insertion (0.4664).
- No clear improvement over the best baseline (Clarke-Wright).

Runtime

- Nearest Neighbor is fastest (~43 ms), Clarke-Wright is moderate (~79 ms).
- The evolved solver runs slower (~161 ms) than these baselines but much faster than Regret Insertion (~1472 ms).

Robustness

- The evolved solver exhibits low robustness dispersion (0.0035), indicating consistent performance.
- Clarke-Wright has slightly higher dispersion (0.0078).
- Regret Insertion shows high dispersion (0.0869), indicating less consistent results.

Code Novelty and Complexity

- Evolved solver shows substantial code novelty (0.657) and some complexity (12.1), indicating significant changes compared to fixed baselines with zero novelty.

Conclusion

- The evolved solver **did not clearly beat** the best baseline (Clarke-Wright Savings) in terms of objective gap or runtime.

- It demonstrated competitive feasibility and robustness with moderate objective gap and runtime, exceeding Nearest Neighbor performance for objective but not surpassing Clarke-Wright.
- Overall, the evolved solver offers a balanced trade-off but not a definitive improvement over the strongest fixed baseline.